

DAY 2 – DIAGNOSING HARDWARE FAILURES

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Course: Diploma in Cyber Security

Topic: Diagnosing Hardware Failures

TASK 1 – COMMON HARDWARE ISSUES

1. RAM Problem

Issue: RAM is loose or damaged.

Symptom: The computer may not start or the screen may show nothing. Sometimes it can also make beep sounds.

2. Power Supply Problem

Issue: The power supply is not working properly.

Symptom: The PC may not turn on or it may suddenly switch off.

3. HDD / SSD Problem

Issue: The storage drive may be damaged or have a problem.

Symptom: The computer can become slow or may not start properly.

4. Overheating

Issue: The CPU or another part of the computer gets too hot.

Symptom: The PC may become slow, freeze, or suddenly shut down.

5. Graphics Card Problem

Issue: The graphics card may not be connected properly or may have a fault.

Symptom: The monitor may show no display or the screen may show strange lines or graphics.

TASK 2 – LAPTOP NOT CHARGING

Problem:

The laptop is not charging even when the charger is connected.

Steps I would follow:

1. Check the Power Socket

First, I would check if the wall socket is working by trying another device.

2. Check the Charger

I would check the charger, cable and adapter for any visible damage.

3. Check the Charging Port

I would check the laptop charging port for dust, damage or a loose connection.

4. Check the Charger Voltage

I would use a multimeter to check the output voltage of the charger and compare it with the voltage written on the charger.

5. Check the Battery

If the battery is removable, I would check if it is properly connected.

6. Try Another Compatible Charger

If I have another compatible charger, I would try it to see if the laptop starts charging.

Result:

If the charger is not giving the correct voltage, the charger may be faulty. If the charger is fine, the problem could be with the charging port or battery.

TASK 3 – TESTING A USB CHARGER

The normal USB charger output is around **5V**. The expected range given in the task is **4.75V to 5.25V**.

Steps:

1. Set the multimeter to **DC Voltage**.
2. Connect the black probe to the **negative/GND** side.
3. Connect the red probe to the **positive/VBUS** side.
4. Check the number shown on the multimeter.
5. Write down the measured voltage.
6. Compare it with **4.75V – 5.25V**.

Example:

Measured Voltage: 5.02V

Expected Range: 4.75V – 5.25V

Result:

The measured voltage is within the expected range, so the charger output looks normal.

Note: If you actually use a multimeter or simulator, replace **5.02V** with your own reading.

TASK 4 – TESTING THE 12V RAIL

Problem:

The PC is randomly shutting down.

I would check the **12V rail** of the ATX connector using a multimeter.

Steps:

1. Turn off the PC and disconnect it from the power.
2. Disconnect the ATX connector from the motherboard.
3. Set the multimeter to **DC Voltage**.
4. Find the **yellow wire** and **black wire**.
5. Put the black probe on the black wire.
6. Put the red probe on the yellow wire.
7. Check the voltage shown on the multimeter.
8. Write down the reading.

Example:

Measured Voltage: 12.08V

The reading is above **11.5V**, so it does not indicate the low-voltage condition mentioned in the task.

If the voltage is below 11.5V:

It could mean that the **power supply is weak or faulty**. This may cause the PC to become unstable or shut down randomly.

CONCLUSION

In this assignment, I learned about some common **hardware problems and their symptoms**.

I learned how to check problems related to **RAM, power supply, storage, overheating and graphics cards**. I also learned the basic steps to check why a laptop is not charging.

I learned how a **multimeter** can be used to check the voltage of a USB charger and the 12V rail of an ATX connector.

Overall, this task helped me understand how basic hardware problems can be checked step-by-step using simple tools.